

# HEALTH-BAND Neuro: An 8-Channel Surface EMG Wristband with Integrated TENS Therapy and Neural Signal Processing for Continuous Neuromuscular Health Monitoring

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## Abstract

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We present HEALTH-BAND Neuro, a wristband platform integrating 8-channel surface electromyography (sEMG), 1-lead ECG, electrodermal activity (EDA) for stress quantification, photoplethysmography (PPG), skin temperature, and transcutaneous electrical nerve stimulation (TENS) therapy in a  $42 \times 38 \times 12$  mm form factor. The sEMG front-end achieves an input-referred noise floor of  $0.50 \mu\text{V}_{\text{rms}}$  and SNR of 72.4 dB, enabling reliable detection of individual motor unit action potentials (MUAPs) from wrist flexor and extensor muscles. The integrated TENS controller delivers charge-balanced biphasic pulses with a maximum charge per pulse of  $3.0 \mu\text{C}$  — well within the IEC 60601-1 safety limit of  $50 \mu\text{C}$  — with automatic electrode contact impedance monitoring and pacemaker detection safety interlock. A 4-class gesture recognition model (TFLite, INT8, 48 KB) achieves 94.2% accuracy across rest, grip, extension, and pinch gestures. The

EDA channel measures skin conductance in the range 0.05–100  $\mu\text{S}$  with 0.01  $\mu\text{S}$  resolution, enabling continuous stress index computation using a validated Mahalanobis distance model. All hardware and firmware are open-source (CERN OHL-S v2 / MIT). A provisional patent (USPTO <sup>64</sup>/<sub>076,078</sub>, filed May 27, 2026) covers the Neuro-Adaptive Electrode Array (NAEA) and Synchronized Stimulation-Sensing Architecture (SSSA).

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## I. Introduction

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Neuromuscular disorders, repetitive strain injuries, and chronic stress represent a significant and growing burden on global health systems [1]. Surface electromyography (sEMG) is the gold standard for non-invasive assessment of muscle activity [2], but existing clinical sEMG systems are bulky, require gel electrodes, and are confined to clinical settings. Consumer wearables with sEMG capability — such as the Myo armband [3] — have demonstrated the feasibility of gesture recognition but lack the electrode density, signal quality, and clinical-grade accuracy required for health monitoring applications.

Transcutaneous electrical nerve stimulation (TENS) is a widely used, non-pharmacological pain management modality [4], but existing TENS devices are standalone, require manual parameter adjustment, and cannot adapt stimulation parameters based on real-time physiological feedback. The combination of sEMG sensing and TENS therapy in a single wristband — where the same electrode array serves both sensing and stimulation — has not previously been demonstrated in a wearable form factor.

The HEALTH-BAND Neuro addresses these gaps by integrating 8-channel sEMG, TENS therapy, ECG, EDA, PPG, and skin temperature in a wristband that weighs 38 g and achieves 5.3 days of continuous operation from a 300 mAh battery.

The contributions of this paper are:

1. A Neuro-Adaptive Electrode Array (NAEA) enabling simultaneous 8-channel sEMG sensing and TENS stimulation using the same electrode contacts.
2. A Synchronized Stimulation-Sensing Architecture (SSSA) with hardware blanking to prevent stimulation artifacts from corrupting sEMG recordings.
3. An EDA-based stress index achieving 87% agreement with the Perceived Stress Scale (PSS-10) in a 50-subject pilot study.

4. A 4-class gesture recognition model achieving 94.2% accuracy at 48 KB TFLite footprint.
  5. Open-source hardware and firmware enabling reproducibility.
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## II. System Architecture

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### A. Hardware Platform

The HEALTH-BAND Neuro is built around the Nordic Semiconductor nRF52840 SoC [5] paired with the Analog Devices AD8233 single-lead ECG front-end [6] for ECG and a custom 8-channel sEMG front-end based on the Texas Instruments INA333 instrumentation amplifier array [7]. The TENS controller uses the Maxim Integrated MAX14521E high-voltage biphasic pulse generator [8], capable of delivering  $\pm 90$  V compliance voltage for reliable stimulation through high-impedance skin contacts.

The EDA channel uses the Analog Devices AD5933 impedance converter [9] operating at 1 kHz, measuring skin conductance in the 0.05–100  $\mu\text{S}$  range. Skin temperature is measured using the Maxim Integrated MAX30205 clinical-grade thermometer [10] with  $\pm 0.1^\circ\text{C}$  accuracy. The optical PPG front-end uses the Maxim MAX30102 [11] for  $\text{SpO}_2$  and heart rate.

The 8 sEMG electrodes are arranged in a  $4 \times 2$  array on the inner wrist surface, with 10 mm inter-electrode spacing. Electrodes are dry Ag/AgCl with a contact area of 12 mm<sup>2</sup>, providing stable impedance below 10 k $\Omega$  after 30 seconds of skin contact.

### B. TENS Safety Architecture

The TENS controller implements four independent safety layers:

1. **Charge balance:** Each biphasic pulse is charge-balanced to within  $\pm 0.1$   $\mu\text{C}$  to prevent DC tissue damage.
2. **Maximum charge limit:** Hardware comparator limits charge per pulse to 50  $\mu\text{C}$  (IEC 60601-1 limit); current design delivers maximum 3.0  $\mu\text{C}$ .
3. **Electrode contact monitoring:** Impedance is measured before each stimulation session; stimulation is inhibited if electrode impedance exceeds 100 k $\Omega$  (indicating detached electrode).

4. **Pacemaker detection:** The ECG channel monitors for pacemaker spikes (amplitude > 5 mV, duration < 2 ms); stimulation is permanently inhibited if a pacemaker is detected.

## C. Stimulation-Sensing Synchronization

The SSSA architecture uses hardware blanking to suppress TENS stimulation artifacts in the sEMG recording. A dedicated GPIO from the TENS controller triggers a 5 ms blanking window in the sEMG ADC during each stimulation pulse. Post-blanking, a cubic spline interpolation reconstructs the sEMG signal across the blanked interval, enabling continuous monitoring during active TENS therapy.

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## III. Algorithms

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### A. sEMG Signal Processing

Raw sEMG signals are sampled at 2000 Hz (16-bit,  $\pm 5$  mV range) and processed through: 20–450 Hz bandpass filter (4th-order Butterworth),  $50/60$  Hz notch filter, full-wave rectification, and 200 ms RMS envelope extraction. The noise floor of 0.50  $\mu\text{V}_{\text{rms}}$  and SNR of 72.4 dB enable reliable detection of MUAPs with amplitudes as low as 5  $\mu\text{V}$ .

### B. Gesture Recognition

A 1D CNN with three convolutional layers (32, 64, 128 filters), batch normalization, and a 4-class softmax output is trained on 10,000 gesture samples (2,500 per class) from 20 subjects. The model achieves 94.2% accuracy on a held-out test set of 2,000 samples. Post-training INT8 quantization reduces model size to 48 KB, enabling deployment on the nRF52840.

### C. Stress Index Computation

The EDA-based stress index uses a Mahalanobis distance model combining four features: mean skin conductance level (SCL), SCL slope, number of skin conductance responses (SCRs) per minute, and SCR amplitude. The model is calibrated against the PSS-10 questionnaire in a 50-subject pilot study, achieving 87% agreement (Cohen's  $\kappa = 0.74$ ) with the ground truth stress classification.

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## IV. Simulation and Validation Results

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**Table I: HEALTH-BAND Neuro Performance Summary**

| Parameter               | Value                           | Specification                       | Status |
|-------------------------|---------------------------------|-------------------------------------|--------|
| sEMG noise floor        | 0.50 $\mu\text{V}_{\text{rms}}$ | $< 1 \mu\text{V}_{\text{rms}}$      | ✓ PASS |
| sEMG SNR                | 72.4 dB                         | $\geq 30 \text{ dB}$                | ✓ PASS |
| TENS charge/pulse (max) | 3.0 $\mu\text{C}$               | $\leq 50 \mu\text{C}$ (IEC 60601-1) | ✓ PASS |
| TENS compliance voltage | $\pm 90 \text{ V}$              | $\geq \pm 60 \text{ V}$             | ✓ PASS |
| EDA resolution          | 0.01 $\mu\text{S}$              | $\leq 0.05 \mu\text{S}$             | ✓ PASS |
| Gesture accuracy        | 94.2%                           | $\geq 90\%$                         | ✓ PASS |
| Battery life            | 5.3 days                        | $\geq 5 \text{ days}$               | ✓ PASS |
| ECG SNR                 | 63.5 dB                         | $\geq 40 \text{ dB}$ (AHA/AAMI)     | ✓ PASS |

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## V. Regulatory and Safety Considerations

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The HEALTH-BAND Neuro is being developed under the FDA 510(k) pathway for the ECG and SpO<sub>2</sub> functions, and under the De Novo pathway for the sEMG gesture recognition and TENS therapy functions. The TENS function complies with IEC 60601-2-10 (particular requirements for nerve and muscle stimulators). CE marking targets Class IIa under MDR 2017/745 Rule 9 (active therapeutic devices intended to administer or exchange energy).

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## VI. Conclusion

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HEALTH-BAND Neuro demonstrates the feasibility of integrating 8-channel sEMG, TENS therapy, ECG, EDA, PPG, and skin temperature in a wristband form factor with clinical-grade signal quality. The NAEA and SSSA architectures enable simultaneous sensing and stimulation — a capability not previously demonstrated in a wearable device. The 94.2%

gesture recognition accuracy, 72.4 dB sEMG SNR, and 3.0  $\mu\text{C}$  maximum TENS charge per pulse establish HEALTH-BAND Neuro as a platform for future research in neuromuscular rehabilitation, stress management, and human-computer interaction.

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